BECOMING AN AI-AUGMENTED ANALYST

A five-stage workflow that turns technical reporting into decision support

This deck presents the process.
Worked examples with real findings will be shown separately.

Mark Ciganovic

THE TECHNICAL WORKFLOW WAS STRONG — BUT INCOMPLETE

Reporting alone is not decision support

OLD REPORTING WORKFLOW

Metric request
→ SQL extraction
→ R processing
→ Report or dashboard

Missing: the decision, the design,
and a defensible next action.

WHAT WAS MISSING

The decision behind the request
A deliberate analysis design
A defensible interpretation
A specific next action

THE CORE LOGIC

Decision → Hypothesis → Measurement → Evidence → Action

01

Start

Surface possible decisions

02

Framing

Lock one precise decision

03

Design

Hypothesis + measurement plan

04

Execution

SQL + R evidence production

05

Finish

Interpretation + recommended action

In my workflow, Stage 3 provides the greatest leverage for AI assistance.

STAGES 1 & 2 — GET THE DECISION

Start by clarifying the real need

01 START

Do not treat the requested metric as the real need.

Offer two or three concrete business decisions the metric could support, then invite the stakeholder to choose which one matters most.

02 FRAMING

Turn the response into one precise decision.

Ask one short clarifying question that identifies the action, priority, or resource choice they are trying to make.

WHEN THE CONVERSATION GETS STUCK

Gentle ways to keep moving toward a clear decision

When the answer stays vague

Gently narrow the options

“Would it help if we focused on whether to [A] or [B]? Or is there a different decision that would be more useful?”

When they prefer just to see the numbers

Acknowledge the preference and offer a practical path

“Happy to pull the numbers. To make sure they’re useful, would you like me to use [X] as a working decision for now, or pause until the priority is clearer?”

When multiple stakeholders are involved

Clarify whose decision the analysis should support

“There may be several perspectives here. For this piece of work, which decision would you like it to help with?”

A SIMPLE GUARDRAIL

Decision or documented non-decision

After a couple of clarifying attempts:

- Proceed with an explicitly stated working decision, or
- Pause and note that a clear decision was not yet available.

This keeps the work useful and avoids analysis that has no decision to support.

STAGE 3 — DESIGN THE TEST

Convert the decision into a testable hypothesis and measurement plan

Hypothesis

- One clear, testable claim
- Tied directly to the decision
- Can be tested and potentially falsified

Primary KPI

- Exact numerator & denominator
- Population & time window
- Unit of analysis / grain

Controls

- Key segments
- Major confounders
- What must be ruled out

STAGE 4 — EXECUTION

Produce evidence with modular R code and AI-assisted quality control

TECHNICAL CORE

SQL extraction and joins
R / Tidyverse processing
Interpretable modeling
Tables, charts, Excel reports, and dashboards

AI-ASSISTED CONTROL LAYER

Review grain and denominators
Check joins and data quality
Challenge assumptions
Validate against the design

STAGE 5 — FINISH

Connect evidence to a proportionate action

EVIDENCE DISCIPLINE

State what the results support
State what they do not support
Separate association from causation
Surface remaining uncertainty

DECISION OUTPUT

Recommend the next action
Match confidence to evidence
Identify what to monitor
Define the next question

PROMPT SUMMARY — ONE PAGE

Prompts for Stages 1–3 | Stage 4 = modular R + code review

STAGE 1 — START

Stakeholder sent this request: “[paste]”
Write a short professional reply that:
• Acknowledges the requested metric
• Offers 2–3 concrete business decisions
• Ends by asking which decision they are trying to make
Return only the message.

STAGE 2 — FRAMING

Stakeholder’s decision statement: “[paste]”
Write a short reply that:
• Briefly acknowledges what they said
• Asks one precise clarifying question
• Offers up to three realistic actions if helpful
Return only the message.

STAGE 3 — DESIGN (highest leverage in my workflow)

Decision: “[paste]” → Design before any SQL: (1) Testable hypothesis (2) Primary KPI with exact numerator, denominator, population, time window (3) Unit of analysis (4) Decision threshold (5) Key segments (6) Major confounders (7) One measurement problem to rule out.

STAGE 5 — FINISH

Decision + Design + Findings → Decision-ready interpretation:
(1) Direct answer (2) Strongest evidence (3) What the evidence does not establish (4) Most important caveat (5) Specific recommended action (6) What to measure next.
Distinguish observation from inference. Do not overclaim.

THE PROFESSIONAL SHIFT

Not faster reporting.
Better decisions grounded in transparent evidence.

Technical execution + Analytical design + Disciplined interpretation

AI supports structure and quality control.
Human judgment owns the evidence, interpretation, and recommendation.

MARK CIGANOVIC

AI-Augmented Analyst | Data & Decision Support

linkedin.com/in/mark-ciganovic • github.com/markjamesc

Workflow arc: Decision → Design → Evidence → Action

Technical execution + Analytical design + Disciplined interpretation

Human judgment owns the evidence, interpretation, and recommendation.